GURU KARUPPASAMY G

AI & Full-Stack Developer | Computer Science Engineering Student
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SUMMARY

Computer Science Engineering student at VIT Vellore who builds practical software end to end, from the algorithm underneath to the interface people use. Experience spans AI/ML research, full-stack web applications and computer vision systems, including a hackathon-finalist AI tutor and a deployed budget-optimization tool.

EDUCATION

VIT, Vellore – M.Tech (Integrated) Computer Science Engineering 2024 – 2029
CGPA: 8.47 • 100% Scholarship under the VIT STARS Scheme
GHSS, Rengasamudram – HSC, Science Stream • 91.2% 2022 – 2024

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, HTML, CSS, Servlet, JSP
Frameworks & Libraries: React, Vite, FastAPI, OpenCV
AI & ML: Google Gemini API, ConvNeXt, YOLO, Computer Vision, Deep Learning
Software Engineering: OOP, Data Structures & Algorithms, Dynamic Programming, SDLC, REST APIs, JSON, Debugging, Testing
Databases & Cloud: SQL, Cloud **Networking:**
CAN, TCP/IP, UDP, IPv4, IPv6 **Tools:** VS Code,
Linux Terminal, APIs, Netlify
Design & Media: Graphic Design, Video Editing
Soft Skills: Leadership, Teamwork, Communication, Time Management, Problem Solving

PROJECTS

SMARTGOV – AI-Powered Budget Planner Live Demo | React, FastAPI, Python, Google Gemini, Dynamic Programming

- Built a full-stack AI budget planner that extracts budget, department, project, cost and priority data from uploaded images using Google Gemini.
- Developed FastAPI REST APIs and a Dynamic Programming optimizer that selects high-priority and compulsory projects within a defined budget.
- Designed and deployed a responsive React interface covering AI extraction, budget optimization, project selection and allocation analysis.

Textile-Adaptive Multi-Scale ConvNeXt | Deep Learning, ConvNeXt, Computer Vision

- Developed and evaluated a textile-aware image-classification architecture on pretrained ConvNeXt-Tiny across four textile classes, as part of an academic research team.
- Applied multi-scale feature aggregation, learnable gated fusion and auxiliary branch supervision, reaching 98.13% test accuracy and 0.9812 Macro-F1 (96.33% ± 0.96% via 5-fold cross-validation).
- Ongoing research work.

LUMINA – AI Socratic Tutor Live Demo | React, TypeScript, Vite, Google Gemini API

- Finalist, AI for Inclusive Education (VIT Yantra – Shark Tech). Built and deployed, as part of a team, an AI tutor that guides learners through reasoning rather than giving direct solutions.
- Integrated the Gemini API with adaptive prompting for personalized explanations; implemented authentication, onboarding and streaming AI responses.

Automated Video Summarization | Python, OpenCV, YOLO, FastAPI

- Developed, with a team, a video summarization system combining motion detection, object detection and adaptive keyframe selection.
- Achieved an 89.46% reduction in review time, condensing a 43-minute video into 4 minutes 32 seconds.

ACHIEVEMENTS

- 1st Rank in the District, Class 12 Board Exams; 2nd Rank in the District, TTSE Competitive Examination.
- 100% Scholarship under the VIT STARS Scheme; selected as “One-Day Headmaster” for academic excellence and leadership.

CERTIFICATIONS

- Claude AI in 90 Minutes Productivity Course: Build Your AI Work Assistant – GUVI x HCL
- Mobile App Development Workshop – VIT • Hardware & Networking Course – Kalai Tally Institute • Internshala Student Partner Program